"EVALUATING THE IMPACT OF QR-BASED DIGITAL LENDING ON MICRO RETAILER GROWTH: A GROUNDED APPROACH"

A Dissertation submitted to the Department of Commerce in fulfilment of the requirements for the Degree of

Bachelor of Commerce

By

AKASH SARASWAT

University Roll No: 22065503284
College Roll No.: 2022/2287

Under the Guidance of

Ms. Seema Choudhary

Department of Commerce





SHAHEED BHAGAT SINGH EVENING COLLEGE
UNIVERSITY OF DELHI
MAY 2026

CERTIFICATE

This is to certify that **Akash Saraswat** *University Roll number* ***22065503284*** has successfully completed the 4th year dissertation titled ***"Evaluating the impact of QR-based digital lending on micro retailer growth: A grounded approach"*** under the supervision of **Ms. Seema Choudhary** in the partial fulfilment of the requirement for *Bachelor of Commerce (Department of Commerce)*, Shaheed Bhagat Singh Evening College, University of Delhi.

And to the best of my knowledge this dissertation has not been submitted in part or full to any other University or Institution for the award of any degree or diploma.

Ms. Seema Choudhary
Supervisor
Department of Commerce
Shaheed Bhagat Singh Evening College
University of Delhi

Dr. Suman Swami
Co-supervisor
Department of Hindi
Shaheed Bhagat Singh Evening College
University of Delhi

CONFERENCE CERTIFICATE



DEPARTMENT OF COMMERCE
RAM LAL ANAND COLLEGE
UNIVERSITY OF DELHI
1st International Conference on Sustainability, Resilience and Inclusive Development
(ICSRD–2026)
on
9-10 APRIL, 2026

CERTIFICATE
OF PRESENTATION

Conferred upon (Professor/ Associate Professor/ Assistant Professor/ Research Scholar/ UG/ PG) Akash Saraswat
for his/ her paper titled Evaluating the impact of QR-based digital lending on micro retailer growth : A grounded approach
at 1st International Conference on Sustainability, Resilience and Inclusive Development organized by Department of Commerce, Ram Lal Anand College, University of Delhi

MR. RAJINDER SINGH
Convenor &
Teacher in Charge

DR.PREETI DEVI
Co-Convenor

PROF. RAKESH KUMAR GUPTA
Principal



DEPARTMENT OF COMMERCE
DYAL SINGH COLLEGE
UNIVERSITY OF DELHI

CERTIFICATE
OF PRESENTATION

This is to certify that

Akash Saraswat

has successfully presented a research paper titled

EVALUATING THE IMPACT OF QR-BASED DIGITAL LENDING ON MICRO RETAILER GROWTH: A GROUNDED APPROACH

at the NATIONAL CONFERENCE ON SUSTAINABLE & INCLUSIVE COMMERCE : AI INNOVATIONS FOR VIKSIT BHARAT @ 2047, held on 7 April 2026.

We appreciate their valuable contribution to academic research and scholarly discussion.

Asso. Prof. Jyotsna Khaitan
Teacher In-Charge

Prof. V.K. Paliwal
Principal

ACKNOWLEDGMENT

I would like to express my sincere gratitude to my research supervisor, **Ms. Seema Choudhary**, Assistant Professor, Department of Commerce, Shaheed Bhagat Singh Evening College, University of Delhi, for her valuable guidance, constant support, and encouragement throughout the completion of this research study. Her suggestions, motivation, and approachable nature helped me at every stage of the research process. I am truly thankful for the time and effort she invested in guiding me whenever I faced difficulties during this study.

I would also like to thank **Shaheed Bhagat Singh Evening College, University of Delhi,** for providing me with the opportunity and academic environment necessary to conduct this research work successfully.

I express my heartfelt gratitude to all the respondents, shop owners, and micro-retailers who participated in the field survey and shared their valuable experiences and opinions related to fintech lending and digital payments. Their cooperation played an important role in the completion of this study.

I am also thankful to my teachers, classmates, friends, and family members for their continuous support, encouragement, and motivation throughout this research journey.

Lastly, I thank everyone who directly or indirectly contributed to the successful completion of this research paper.

Akash Saraswat
B.Com. (Honours)
Shaheed Bhagat Singh Evening College, University of Delhi
Date: 12-05-2026

TABLE OF CONTENTS

Contents	Page No.
Certificate	ii
Conference Certificate	iii
Acknowledgment	iv
Table of Contents	v
List of Tables	vi
List of Figures	vii
List of Abbreviations	viii
Abstract	ix
Introduction	1
Literature Review	2
Objectives of the Study	8
Research Questions	8
Hypotheses	9
Research Methodology	9
Limitations of the Study	10
Research Findings	11
Evaluation of Hypotheses	16
Discussion	17
Recommendations	19
Conclusion	20
Appendices	21
References	24

LIST OF TABLES

Table No.	Title	Page No.
Table 1	Factor affecting QR-Lending	4
Table 2	Objectives of Study	8
Table 3	Research Questions	8
Table 4	Comparison of Interest Rates across Different Lending Channels for MSMEs	17
Table 5	Illustrative Example of Loan Cost	18

LIST OF FIGURES

Figure No.	Title	Page No.
Figure 1	PRISMA Diagram	5
Figure 2	Documents by year	6
Figure 2	Documents per Year by Source	6
Figure 3	Documents by Author	6
Figure 4	Documents by Affiliation	6
Figure 5	Documents by Country/Territory	7
Figure 6	Documents by Type	7
Figure 7	Documents by Subject Area	7
Figure 8	Documents by Funding Sponsor	7
Figure 10	Fintech Loan Offer Reach among Respondents	11
Figure 11	Adoption of Fintech Loans among Respondents	12
Figure 12	Interest Rate Perception among Respondents	12
Figure 13	Impact of Fintech Loans on Business Performance	13
Figure 14	Financial Literacy Level of Respondents	14

LIST OF ABBREVIATIONS

MSME - Micro, Small and Medium Enterprises

RBI - Reserve Bank of India

QR - Quick Response Code

NBFC - Non-Banking Financial Company

GDP - Gross Domestic Product

POS - Point of Sale

ABSTRACT

India has emerged as one of the leading fintech economies in the world, largely driven by the rapid adoption of digital payment technologies in everyday transactions. Since the onset of the COVID-19 pandemic, an increased adoption of QR payments has resulted in the gradual shift away from cash transactions. This is due to the ease of making digital payments using QR technology and creating digital records of transactions on behalf of the merchants. Based on these digital transaction records, various fintech firms are currently providing collateral-free digital loans, claiming that this helps enhance the growth of their businesses and improve financial inclusion within the MSME sector.

The motivation behind this study emerged from observing that many small shopkeepers and micro-retailers frequently require financial support either for business-related needs or personal emergencies. At the same time, fintech companies are increasingly promoting QR-linked digital loans through platforms such as Paytm, BharatPe, and Mobikwik. This created a need to examine whether such loans are actually helping the merchants on the ground and how the retailers view these lending services in practice.

Existing studies on fintech lending have largely relied on secondary data and company reports, while relatively limited field-based research has focused specifically on QR-based lending models among Indian micro-retailers. Earlier studies have also given limited attention to merchants' financial literacy, awareness of government schemes, understanding of loan terms, repayment conditions, and borrowing behaviour. This study attempts to address these gaps through direct field-based investigation.

The research relies on primary data collected through questionnaires and direct interactions with 52 micro-retailers in Delhi, including kirana stores, chemist shops, hardware stores, and general convenience shops using QR-based payment systems. The study examines merchants' experiences, perceptions, borrowing behaviour, financial literacy, and awareness regarding fintech lending and government credit schemes.

The findings of this study show a clear gap between the availability of fintech loans and their actual utilization among the merchants. Despite the popularity of QR code payment systems, most businesses are unwilling to consider digital lending facilities due to high interest rates, repayment pressures, lack of clarity, and lack of awareness about the lending process. Many participants indicated that borrowing money through informal means was better than using fintech platforms for loan purposes. Another discovery was that the merchants do not have adequate understanding of financial terms, repayment policies, and government credit schemes.

In conclusion, this research provides useful insight into the working of QR code-based FinTech lending in actual practice in India. Fintech lending could be beneficial to financial

inclusion as well as small-scale businesses, but the problem lies in financial literacy, communication gaps, limited awareness, and trust-related concerns which are continuously affecting its actual adoption and effectiveness at the ground level.

1. Introduction

Almost everyone is familiar with the term "FinTech," which describes the growing group of technology-based financial services companies–those that deliver a wide range of financial solutions for consumers and merchants alike. While each FinTech platform serves many different types of businesses or segments, the purpose of this study is to explore whether micro-retailers have access to meaningful opportunities via FinTech lending models from merchants. Though FinTech companies commonly cite that they support small businesses through digital credit, it is important to determine if these statements equate to actual benefits at the ground level.

QR codes (Quick Response Codes) are now widely utilized throughout retail locations in India as a result of recent pandemic-related advancements in QR code-based payment systems. They allow merchants to easily receive payments through digital means (Mookerjee et al., 2022).

Beyond facilitating payments, these systems have also opened new avenues for data-driven lending. By analysing transaction data generated through QR payments, FinTech companies have introduced credit facilities for merchants, positioning themselves as an alternative source of business finance for the merchants (*Paytm Annual Report 2025*, 2025).

In India, even with these advancements in technology, many MSMEs (Micro, Small and Medium Enterprises) do not have adequate access to formal credit. Only 14% of the 64 million MSMEs in India have access to formal credit through banks or other financial institutions. This creates a large credit gap, which often has to be filled by informal lenders (Fedbank Financial Services Limited, n.d.). Formal lending institutions (banks and similar regulated financial organizations) are typically able to lend money to borrowers based on their approved criteria for past credit history, collateral requirements and extensive documentation. However, the criteria required for micro and small businesses are often not met. The MSME sector contributes around 29% to India's GDP (Shelly et al., 2020), so better access to financing for these businesses is vital to support India's economic development.

To address this ongoing credit gap and to help small and medium-sized enterprises expand their businesses, fintech companies are offering loans that do not require collateral or fintech products that are not secured. In the overall economic landscape, these trends have developed into a matter of increasing importance, especially while India strives to reach its goal of growing into the world’s third-largest economy where Micro & Small Business Growth & Financial Strengthening is essential (*Vision of a USD 5 Trillion Indian Economy*, 2018). The study aims to conduct research to test the accuracy of claims made by prominent fintech companies regarding the availability of financially viable and growth-oriented credit for small businesses in the Indian micro-retail sector. Even though various fintech platforms promote inclusive lending models to support small business growth, there is still insufficient field-

based data available to substantiate these claims. Much of the prior research relies on secondary data from industry reports or companies, indicating a gap in field-based research that this study investigates through primary data collection.

The study also focuses on financial literacy among micro-retailers, in addition to ease of access to credit. Although India has actively promoted financial inclusion through government initiatives such as the PM MUDRA Yojana and other MSME-focused credit schemes (Nambiar, 2025; Sushmita, 2025), limited financial literacy often restricts merchants' ability to make informed borrowing decisions (*Financial Literacy | RBL Bank*, n.d.)

Understanding loan terms, interest rates, repayment obligations, regulatory rules, and digital security continues to be a challenge for many small retailers. Therefore, this research seeks to examine whether improved access to fintech-based credit is supported by adequate financial awareness, or whether gaps in financial literacy continue to pose risks for micro-retailers.

2. Literature Review

QR, a concept frequently encountered in daily life, has transformed the Indian landscape for small and medium retailers—not only by making transactions smooth and easy, but also by creating opportunities to access business loans at relatively low interest rates, with collateral-free terms, minimal documentation, and without the requirement of a prior credit history. This lending model helps meet the financing needs of India's MSME sector, which contributes around 30% to the country's GDP, underscoring the importance of this segment to the economy (Gugulothu & Aleti, 2024).

Existing research suggests that cashless payment systems, particularly QR-based payments, have made access to credit easier for small businesses. Studies indicate that merchants using digital payments are able to obtain loans with comparatively lower interest rates, higher loan amounts, and a lower probability of default, as digital transaction histories allow lenders to better assess repayment capacity (Ghosh et al., 2025). In many cases, financial technology companies manage loan repayments from merchants by linking instalments directly with the merchant's digital payment transaction account. This facilitates a smoother repayment process and helps in timely recovery of loan amounts.

The literature shows that fintech has improved financial reach for small entrepreneurs, especially those who were previously excluded from traditional banking systems. Many small businesses do not have formal financial records or credit history, which makes it difficult for them to secure loans from banks. Given these challenges, digital payment systems, mobile wallets, and peer-to-peer (P2P) lending platforms have helped such entrepreneurs access financial services without depending on traditional banks. Existing studies also suggest that fintech-based payment systems can support business growth by

making transactions easier and improving customer reach (Dangkeng & Munir, 2025). Unlike traditional banks, fintech lending platforms do not depend on collateral or lengthy paperwork. Instead, they use digital transaction data and basic customer behaviour to assess borrowers. This approach has made it easier for entrepreneurs, especially those in food services, retail, and small digital businesses, to access small loans for needs such as business expansion, managing inventory, and adopting new technologies (Dangkeng & Munir, 2025b). Existing studies suggest that fintech-based payment systems can help increase sales for small businesses, as customers find it easier to make even small digital payments. The convenience of cashless transactions also encourages greater customer participation and smoother purchasing processes (Sharma et al., 2023).

Some international studies also highlight the role of QR-based payment systems in supporting MSME growth. For example, systems like QRIS in Indonesia have helped improve digital payment adoption and business transactions. These findings provide useful insights, although the context may differ from India (Metr, 2024).

Existing studies on digital lending and fintech-based credit for micro-retail entrepreneurs are largely based on secondary data sources such as company-reported transaction records, industry publications, and academic journals. There remains a need for more field-based research examining QR-based digital lending models at the ground level. Most studies rely on data published by fintech companies themselves, which may not fully reflect the actual experiences of micro-retailers. As a result, there is a clear gap in understanding how QR-based loans function in real business environments, whether they genuinely support business growth, and how retailers perceive and utilise these lending models. This study seeks to address this gap by collecting primary data through direct field interactions with micro-retail entrepreneurs.

Although existing literature strongly promotes financial inclusion through digital payments and fintech lending, further research is needed to examine whether micro-retailers adequately understand lending-related terms, schemes, and borrowing conditions associated with digital credit. Most studies focus on access to credit but do not assess retailers' awareness of government schemes, loan terms, repayment conditions, and default policies. This creates a gap between financial inclusion and financial understanding, which requires further examination through field-based primary research.

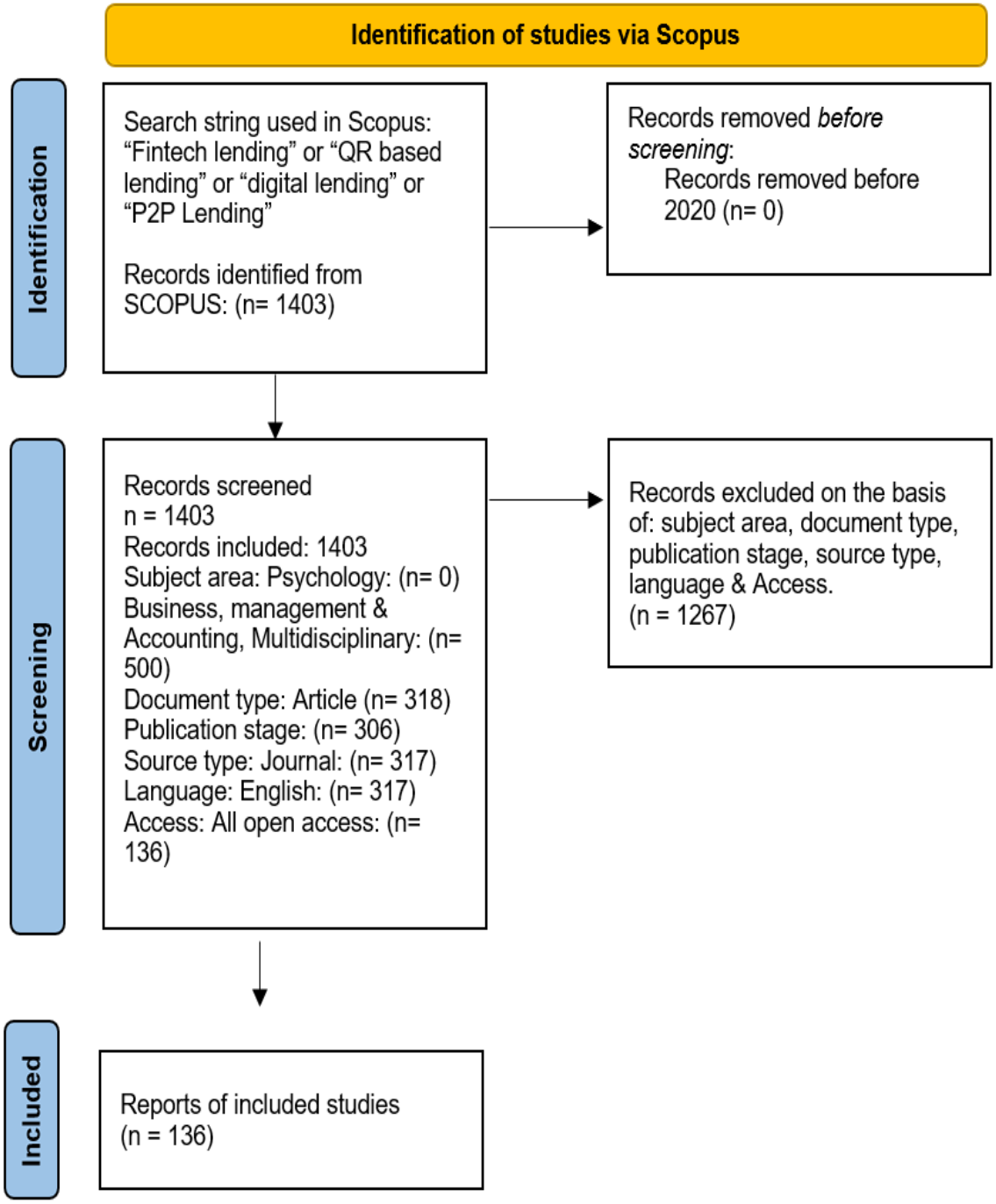
In this research paper, the focus is on examining whether business loans taken by Indian merchants help them grow their businesses, especially at the ground level. While fintech companies claim that their lending models support business growth, field-based evidence examining these claims remains limited but is emerging (Sanjana, 2025). Moreover, this study also explores the level of financial literacy among retailers regarding loan terms, rules

and regulations, and government schemes. This analysis will help in forming a clearer understanding of whether financial inclusion actively promoted in India is translating into informed and effective financial participation among micro and small retailers.

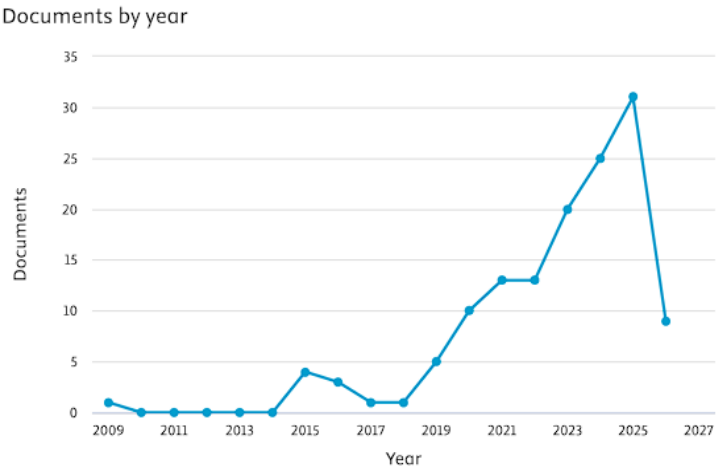
The next section presents the analysis of the reviewed research papers along with the PRISMA diagram used to show how the studies were selected and excluded during the research process.

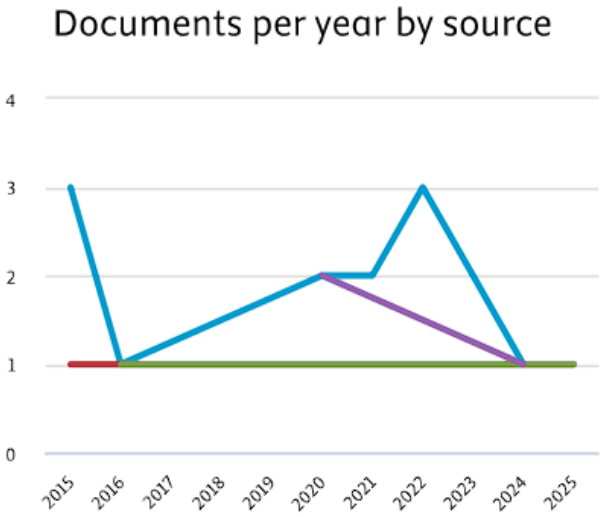
Table 1: Factors affecting QR-Lending

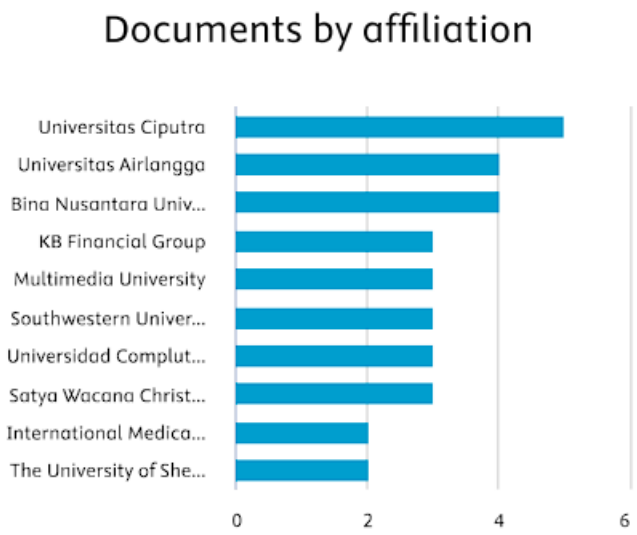
Aspect	QR-Based Lending	Collateral-Free Loans
Key Drivers	Ease of use, trust, performance expectations, literacy	Screening tech, guarantees, digital collateral
Risk Mitigation	Trust in technology, secure platforms	Risk-contingent credit, relationship lending
Target Groups	MSMEs, marginalized vendors	Borrowers lacking tangible assets
Challenges	Adoption barriers, digital literacy gaps	High perceived risks, lack of regulatory clarity


Identification of studies via Scopus
Identification
Search string used in Scopus: "Fintech lending" or "QR based lending" or "digital lending" or "P2P Lending"
Records identified from SCOPUS: (n= 1403)
Records removed *before screening*:
Records removed before 2020 (n= 0)
Screening
Records screened
n = 1403
Records included: 1403
Subject area: Psychology: (n= 0)
Business, management & Accounting, Multidisciplinary: (n= 500)
Document type: Article (n= 318)
Publication stage: (n= 306)
Source type: Journal: (n= 317)
Language: English: (n= 317)
Access: All open access: (n= 136)
Records excluded on the basis of: subject area, document type, publication stage, source type, language & Access.
(n = 1267)
Included
Reports of included studies
(n = 136)

Year	Documents
2026	9
2025	31
2024	25
2023	20
2022	13
2021	13
2020	10
2019	5
2018	1

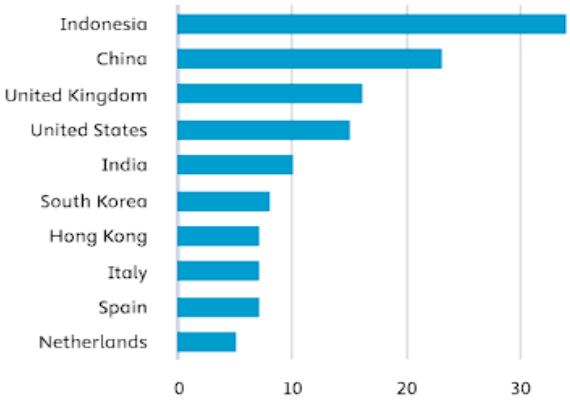




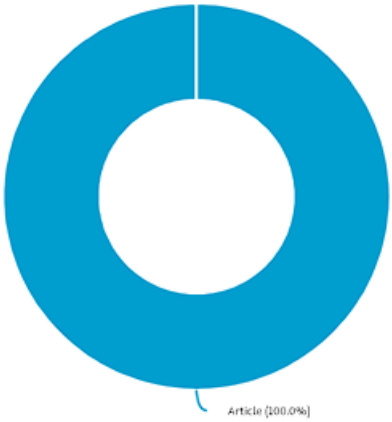




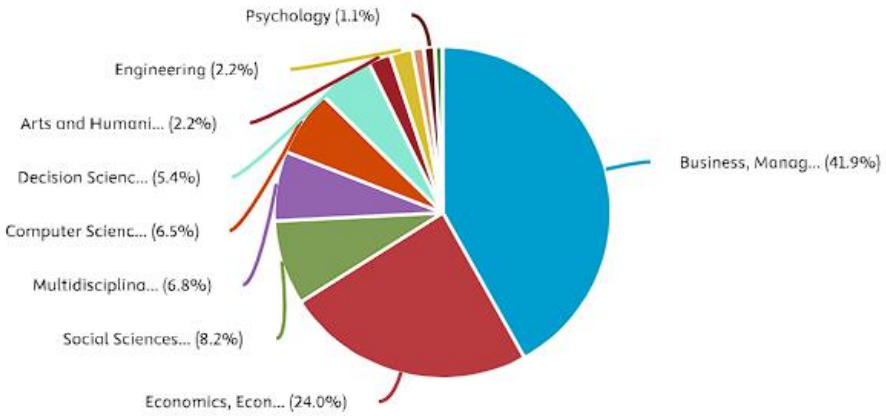
Documents by country/territory



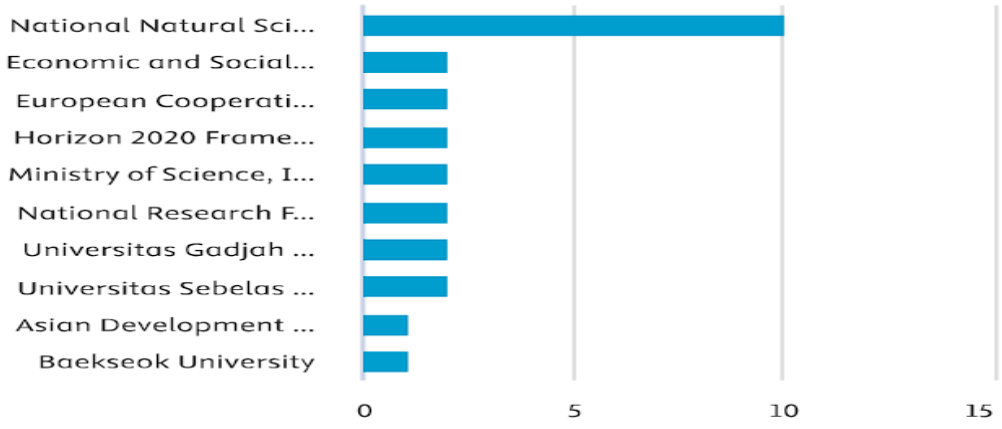
Documents by type



Documents by subject area



Documents by funding sponsor



3. Objectives of the Study

Objective 1	To understand whether QR-based collateral-free loans offered through fintech platforms actually support business growth among micro-retailers in India.
Objective 2	To analyze micro-retailers' experiences and borrowing behaviour, repayment-related issues, challenges, and risk perceptions toward fintech lending platforms.
Objective 3	To examine existing digital lending frameworks, regulatory guidelines, and fintech loan policies related to QR-based microcredit in India.
Objective 4	To evaluate the level of financial literacy among micro-retailers, including their awareness of government schemes and alternative credit options.

Table 2

4. Research Questions

Research Question 1	Do QR-based collateral-free fintech loans support business growth among micro-retailers in India?
Research Question 2	What are the experiences, borrowing behavior, repayment-related issues, and risk perceptions of micro-retailers toward fintech lending platforms?
Research Question 3	How do digital lending frameworks, regulatory guidelines, and fintech loan policies influence QR-based microcredit in India?
Research Question 4	What is the level of financial literacy among micro-retailers regarding government schemes, loan terms, and alternative credit options?

Table 3

5. Hypotheses

Hypothesis 1: Business Growth

Null Hypothesis (H_{01}): QR-based collateral-free digital loans provided by fintech companies do not have a positive impact on the business growth of micro-retailers.

Alternative Hypothesis (H_{11}): QR-based collateral-free digital loans provided by fintech companies have a positive impact on the business growth of micro-retailers.

Hypothesis 2: Financial Literacy

Null Hypothesis (H_{02}): Financial literacy among micro-retailers does not influence their understanding of loan terms, repayment conditions, and awareness of government credit schemes related to digital loans.

Alternative Hypothesis (H_{12}): Higher financial literacy among micro-retailers improves their understanding of loan terms, repayment conditions, and awareness of government credit schemes related to digital loans.

6. Research Methodology

Research Design

The study adopts a "mixed methods research framework", a combination of quantitative and qualitative approaches. This framework is appropriate as the researcher aims to assess the measurable impact of QR-based digital lending on micro-retailer growth, while understanding real-world experiences, perceptions, and behavioural aspects of merchants. The integration of survey data, field interactions, and purposive sampling provides a balanced and practical approach to addressing the research objectives.

Sources of Data

The study uses both primary and secondary data. The researcher gathered field data to support the primary data. Discussions and experiences from the ground level to peer into how QR-based lending is operationalizing, and Such as performance of the business, repayment mechanism, and financial literacy etc. among micro-retailers. Besides, secondary data was called upon to gain an understanding of the RBI guidelines and fintech lending policies.

Primary Data Collection

Primary data was collected through field surveys and structured questionnaires given to micro and small retail shops. The questionnaire primarily consisted of close-ended questions, in addition to some free-response answers on topics like business expansion, loan utilization, The struggle to repay loans, knowledge of government initiatives, the cost of the loan, and general borrowing experience. Furthermore, field interactions helped in the

understanding of retailers' attitudes towards fintech lending, their perception of risks, level of trust in digital platforms, lack of trust in such financial services.

Pilot Testing of the Questionnaire

Before conducting the final survey, a pilot study was done on around 15 micro-retailers to review the questionnaire's clarity and understanding. The pilot survey aided in identifying unclear questions, improving the order of the questions, and ensuring that the questions are in the right order. It was found that there was a good correlation between responses and the study's objectives. On the basis of feedback received, the necessary changes were made. The instrument was finalized based on these modifications.

Sampling Technique

The number of micro-retailers sampled in this study was purposively selected because some of them have been identified as the potential sample population. The study evaluated the participants' knowledge of QR-based fintech lending. This includes those who have either taken out such loans, or looked at loan offers on digital payment platforms. The inclusion criteria required participants to be micro or small retailers who were already using QR-based digital payment systems. The study primarily focused on small retail businesses, including street vendors and other micro-retailers, rather than large businesses, with an emphasis on their business growth and expansion experiences.

Sample Size

The study is based on **52 micro-retailers** from different local markets and areas in Delhi, so that different types of businesses and lending experiences could be included.

Data Analysis Approach

The quantitative data obtained from the questionnaire were analyzed using descriptive statistics, including frequencies, percentages, and comparisons, to assess borrowing patterns, financial literacy, and reported business outcomes among the respondents. The qualitative information obtained through field interactions was interpreted thematically to identify recurring experiences, perceptions, and behavioural patterns related to QR-based lending. The hypotheses were assessed using the descriptive evidence obtained from the survey responses and field observations rather than through inferential statistical significance testing.

Scope of the Study

It examines QR digital lending provided by fintech firms and the experiences of micro-retailers, their financial literacy and problems encountered in digital lending. It also attempts to examine the disconnect between policies' goals and their implementation at the "street level."

7. Limitations of the Study

i) While not all micro-retailers utilize QR-based digital financing loans solely for their businesses, it shows that these loans do not always support the growth of their businesses.

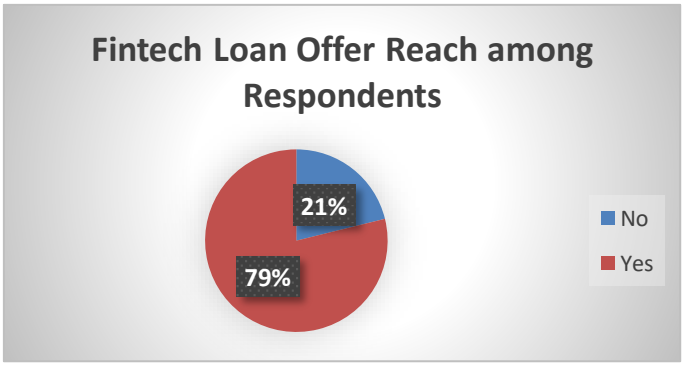
ii) This research does not have access to companies' internal finance records or credit assessment processes from fintech companies and does not contain specific lending data from companies.

iii) This study is limited to micro-retailers in Delhi, and therefore, the findings cannot reflect the experiences of micro-retailers across the country.

iv) There could be inaccuracies in responses since some respondents responded based on what they thought or remembered, or because they were uncomfortable discussing loans and finances.

8. Research Findings

Fintech companies have increasingly expanded access to credit for small businesses, including businesses that may lack traditional collateral or credit histories (Omowole et al., 2024). However, the ground situation should be explored to see if these loans are truly being adopted by merchants and if they aid business stability. The study also evaluates the level of financial literacy by assessing merchants awareness of government schemes such as the Pradhan Mantri Mudra Yojana, Stand Up India Scheme, PM SVANidhi Scheme and key concepts like the Key Fact Statement. The findings of the study are based on 52 survey responses collected from micro-retailers. These findings provide insights into merchants' perceptions of fintech loan offers, interest rates, and their basic financial understanding. Out of the 52 respondents, Figure 1 shows that 79% reported receiving loan offers from fintech companies, while the remaining 21% had not received such offers.

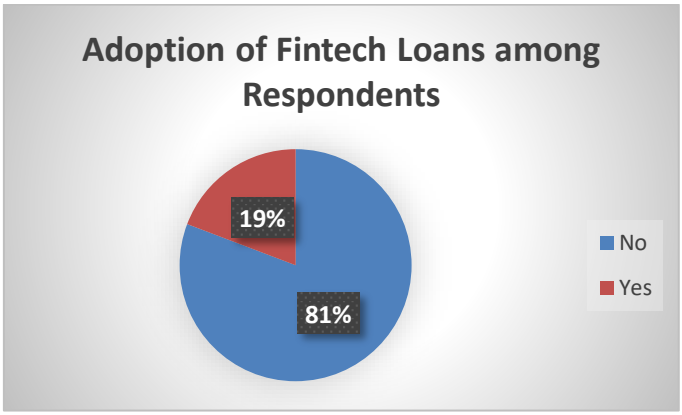


Source: Author's representation

Figure 1: Fintech Loan Offer Reach among Respondents

The respondents who did not receive loan offers were mainly those with lower digital transaction volumes, a higher preference for cash transactions, seasonal business patterns, or

existing loans from other financial institutions. In many cases, merchants had access to QR-based payment systems but used them only in critical situations, while continuing to rely primarily on cash transactions. Additionally, merchants engaged in seasonal businesses or those who had already taken loans from other institutions were less likely to receive loan offers from fintech platforms. These findings suggest that fintech companies are actively attempting to offer loans to a large number of merchants in order to promote financial inclusion. However, this process is influenced by multiple factors such as merchant behaviour, transaction patterns, and business requirements, which fintech platforms may not always be able to accurately assess. At the same time, since loan offers are reaching a majority of merchants, it can be interpreted that fintech platforms are attempting to enhance merchants' purchasing capacity. Nonetheless, this potential does not necessarily translate into actual loan adoption. Now, focusing on the merchants who have taken fintech loans, the data reveals that adoption remains significantly low. Figure 2 clearly shows that 81% of the respondents have not accepted fintech loan offers. This shows that although such loan offers are easily available, a large number of merchants still hesitate to adopt them.



Source: Author's representation

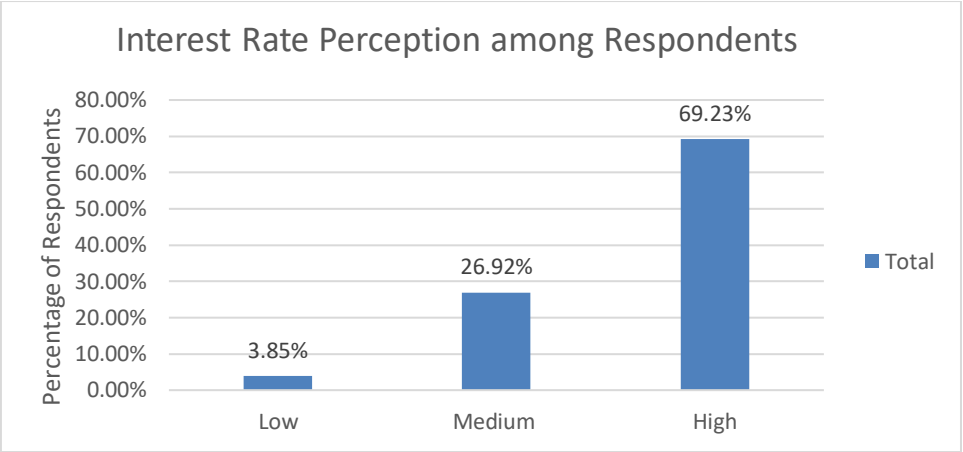
Figure 2: Adoption of Fintech Loans among Respondents

The data clearly show that despite the availability of loan offers, a large proportion of merchants prefer not to take fintech loans due to multiple reasons.

The major reasons for non-adoption include:

1. High interest rates charged on loans

   Figure 3 shows that a significant majority of respondents perceive fintech loan interest rates as high, which acts as a major barrier to the adoption of fintech lending.



Source: Author's representation

Figure 3: Interest rate perception among respondents

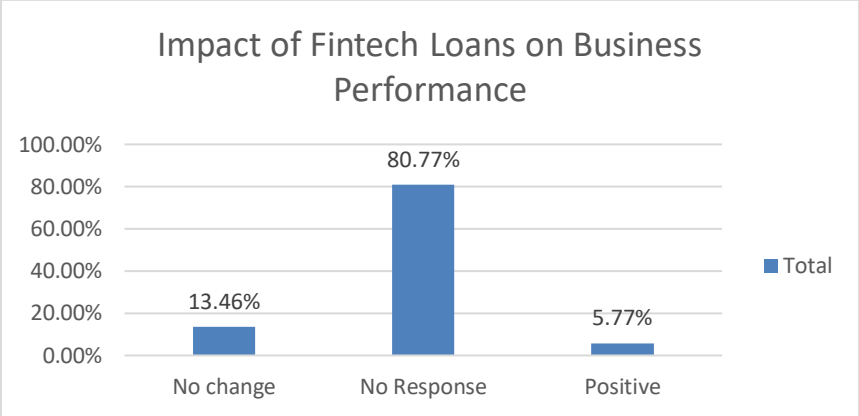
2. Concerns regarding repayment, particularly the impact on daily cash flow in case of default.
3. A significant number of respondents reported a preference for informal lending sources such as personal contacts.
4. Lack of need for credit due to limited business expansion.

These findings indicate that factors such as cost perception, risk, and existing financial practices play a major role in influencing borrowing decisions among micro-retailers. The next important aspect of the study focuses on the financial literacy of micro-retailers. For this, the respondents were asked questions on government schemes and important financial terms. Although India has been working towards achieving financial inclusion on a national level, there is a need for a certain level of financial literacy among merchants, to ensure that they are able to adopt fintech products. The findings reveal that 100% of respondents were unaware of the Pradhan Mantri MUDRA Yojana as a potential source of business financing. This highlights a significant gap between policy initiatives and ground-level awareness.

This point is also evident from an actual case study encountered during the survey: The "Pan Shop Owner" who was physically challenged, along with his wife, showed a keen interest in utilizing the government scheme but had no idea how to do so. In another case, the fruit seller said that in order to avail the government scheme, he needs to make several visits, a process that consumes significant time and makes it difficult for small retailers alongside daily business activities.

These observations indicate that even though financial inclusion initiatives are being promoted

at the policy level, their effective implementation at the ground level remains limited due to lack of awareness and accessibility.



Source: Author's representation

Figure 4: Impact of Fintech Loans on Business Performance

The reported impact of fintech loans on business growth appears limited among the surveyed respondents. Figure 4 shows that most of the retailers did not report any notable improvement in the performance of the businesses after taking fintech loans. However, a few shopkeepers state that these loans are very much helpful during the challenging periods such as monsoon, extreme weather conditions. Particularly for keeping inventory and managing short-term losses some merchants also use these loans to start a small side business. This shows that fintech loans are mostly taken to manage short-term financial needs instead of helping businesses grow in the long run.

A deeper look into the financial literacy results identifies that many of the surveyed respondents demonstrate only minimal background knowledge concerning finances. Most of the respondents would be classified as having low financial literacy as Figure 5 indicates that many of the micro businesses are not yet familiarized with financial concepts and how to borrow.

Many merchants understand that fintech loan products are backed by Non-Banking Financial Companies (NBFCs), and do not require collateral for loan offers. However, they do not have clear knowledge of the terms of those loans and how the repayment works as well as all of the other conditions associated with the loans.

A large portion of merchants who do not understand fintech lending have misconceptions about how it works, such as that they need collateral and/or significant documentation in order to obtain a fintech loan. This lack of knowledge ultimately keeps these merchants from getting loans that would be beneficial for them.

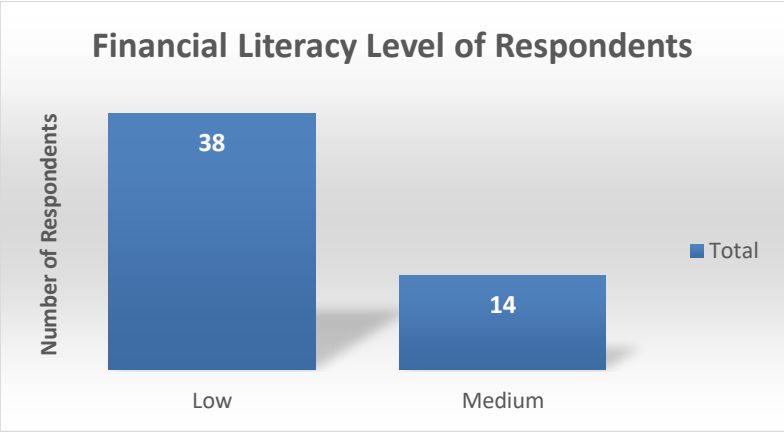


Figure 5: Financial Literacy Level of Respondents

Source: Author's representation

There are also several important field observations identified at the time of survey. Some merchants said that company representatives promote Point of Sale (POS) devices or services as free but later the charges were deducted from their accounts. This reflects a gap between the information provided and actual practices, raising concerns regarding transparency and rules and guidelines.

Similarly, some respondents commented on the strict repayment practices, where even a one-day delay in instalment repayment can lead to quick response from the companies. Sometimes fintech companies put restrictions on access to funds or, in certain cases, temporary control over daily transaction cash flows by the lending platform.

In one case, a vendor who had taken a fintech loan switched to another QR platform without repaying the loan, highlighting potential risks for fintech companies in unsecured lending environments.

Another important observation relates to the difference between the sanctioned loan amount and the actual disbursed amount received by merchants. During field interactions, several respondents reported that the amount credited to their accounts was lower than the total loan amount due to deductions such as processing fees or other charges. However, interest was calculated on the full sanctioned loan amount.

For example, one shop owner reported taking a loan of ₹2,00,000 but received only ₹1,93,000 in his account, while interest was charged on the entire loan amount. A similar situation was observed in another case as well.

This shows a lack of awareness among merchants regarding loan structures and highlights serious concerns related to transparency in digital lending practices.

Difference Between Loan Amount and Disbursal Amount

Many of the merchants taking the loan from fintech highlighted the gap between the loan amount and the disbursal amount. The actual disbursal amount was often lower compared to the amount they were applying for.

It was due to the reduction of the amount due to GST and the processing fee. However, the merchant was unaware of the disbursal amount and thought that the complete amount would be deposited.

Thus, it created confusion regarding the actual cost of borrowing. Despite having it in the company policy, it may not always be communicated clearly and concisely.

POS Device Misrepresentation

Another important aspect highlighted by the merchants was how company representatives sold POS devices to retailers. They were asked to upgrade their existing POS devices such as from 4G to 5G at no monthly or additional cost.

However, many of the merchants have discovered that the monthly cost was being deducted from their accounts for using the POS devices as a service charge. The company must clearly communicate each deduction and information to the merchant so that they are able to make informed decisions without any confusion and miscommunication.

9. Evaluation of Hypotheses

The hypotheses were evaluated based on descriptive evidence from the survey responses and field observations. Since the study did not employ inferential statistical tests, the evaluation indicates whether the observed evidence supports or does not support the proposed relationships rather than establishing statistical significance.

Hypothesis 1 – Business Growth

The descriptive findings do not support H_{11}. Most respondents did not report notable business-growth benefits from QR-based fintech loans, while field interactions indicated that such loans were more commonly used for short-term financial needs, inventory-related expenses, and temporary cash-flow requirements than for major business expansion. Therefore, the positive impact proposed in H_{11} is not supported by the descriptive evidence from the surveyed micro-retailers.

Hypothesis 2 – Financial Literacy

The descriptive findings support H_{12}. The responses and field interactions indicate that limited financial literacy was associated with difficulties in understanding loan terms,

repayment conditions, fees, and government credit schemes. The findings therefore suggest that higher financial literacy is associated with better understanding of digital lending conditions among the surveyed micro-retailers.

10. Discussion

In summary, the research provides the following main insights into QR-based fintech lending services provided to micro-retailers and merchant perceptions. While fintech services have facilitated easier access to credit, their success still depends on factors such as costs, transparency, and the financial awareness of merchants.

1. Comparison of Interest Rates across Lending Sources

A key factor influencing the adoption of fintech loans is the perception of relatively higher interest rates (Haris, 2024). This finding is consistent with Haris (2024), as 69.23% of the respondents in the present study perceived fintech loan interest rates as high. To better understand this, a comparison between traditional bank loans and fintech/NBFC-based loans is presented below.

To examine the variation in interest rates across different lending channels, Table 4 presents a comparative analysis.

Table 4: Comparison of Interest Rates across Different Lending Channels for MSMEs

Lender Category	Typical Interest Rate (%)	Key Characteristics
Public Sector Banks (PSBs)	8% – 11.5%	Lower interest rates; often require collateral and formal documentation.
Private Sector Banks	10% – 20%	Moderate interest rates; depend on credit profile and financial records.
Government Schemes (PMMY, CGTMSE, Mudra)	6% – 10%	Subsidized rates for eligible micro-units; accessed through banks.
NBFCs (Non-Banking Financial Companies)	12% – 30%	Higher rates compared to banks; faster processing; may be unsecured or semi-secured.
Fintech Platforms (QR-based lending)	12% – 24%	Short-term, unsecured loans based on transaction data; quick disbursal.

Source: Author's representation from secondary data.

As shown in Table 4, interest rates associated with fintech and NBFC (Non-Banking Financial Company) lending are comparatively higher than those offered by traditional lending sources. This supports the findings of the present study, where merchants reported "higher borrowing

costs and perceived fintech loans as relatively expensive." Such perceptions may also influence other merchants through informal interactions, which can discourage them from opting for these credit facilities.

2. Transparency and Disbursal Structure

The study also highlights issues related to the transparency of loan structures, particularly the difference between the sanctioned loan amount and the disbursed amount. Many merchants were not fully aware that deductions such as GST and processing fees are applied before the loan amount is credited to their accounts. These findings underscore a gap in "financial awareness and transparency" as merchants often do not clearly understand the structure of digital loans, including the difference between sanctioned and disbursed amounts. This lack of clarity may increase the cost of borrowing, create confusion, and negatively affect trust in fintech lending. In some cases, it may also contribute to financial stress among micro-retailers.

3. Financial Literacy and Decision-Making

The findings of the study indicate that only a small proportion of merchants possess basic financial knowledge, while a majority lack adequate understanding of financial concepts. Many respondents were unable to clearly understand loan terms, repayment structures, and associated charges. Limited financial literacy can lead to poor understanding of loan conditions and often results in unfavourable financial decisions regarding such products, which may increase the risk of financial stress among borrowers. This is evident from the field observations, where merchants were often unaware of government schemes such as PM MUDRA Yojana (2015), Stand-Up India Scheme (2016), and PM SVANidhi scheme.

Moreover, they were unaware of deductions such as GST and processing fees applied before loan disbursement. Interactions with them revealed that they had very limited understanding of key financial aspects such as interest calculations, repayment schedules, and digital loan structures.

Table 5: Illustrative Example of Loan Cost

Component		**Amount (₹)**
Loan Amount (Sanctioned)		2,00,000
Processing Fees & GST (approx.)		7,000
Disbursed Amount (after deductions)		1,93,000
Interest at 18% (average based on observed range from Table 4)		36,000
Total Repayment Amount (sanctioned amount + interest)		**2,36,000**
Total cost of borrowing (fees + interest)		43,000

Source: Author's representation.

On the amount actually received (₹1,93,000), the total cost of ₹43,000 works out to roughly

22.3%, higher than the stated 18% interest rate. This is a simplified calculation that assumes interest for one year on the full sanctioned amount.

This example is illustrative in nature and demonstrates how additional charges and interest can increase the effective cost of borrowing, based on the observed range presented in Table 4. Such situations highlight how low financial literacy can affect borrowing decisions and increase the overall cost burden on micro-retailers. This deficiency limits the ability of merchants to make informed borrowing choices, which may lead to hesitation in adopting fintech loans or result in unfavourable financial outcomes.

4 - Trust and Behavioural Impact (POS Issue)

Field observations related to POS devices reveal important concerns about transparency in "Fintech practices". Many merchants reported that they were told no charges would be deducted after upgrading their POS devices to newer versions (such as 4G or 5G). However, in reality, deductions were still made from their accounts. This creates confusion and shows a form of miscommunication in the way financial products are being presented to the merchants. Such issues need to be resolved by the fintech companies to improve the transparency and build strong trust among users. These findings clearly show a lack of transparency in fintech practices, which can negatively affect trust and long-term adoption among merchants. Trust plays a very important role in financial decision-making, and such experiences may lead merchants to hesitate while adopting fintech lending services. Earlier studies have also noted that the lack of transparency in loan decisions, poor user interfaces, and inadequate customer support can lead to dissatisfaction among users of digital lending platforms (Sekhar & Saheb, 2026). According to (Ramesh et al., 2025), trust in digital payments especially in the informal business settings is deeply rooted and plays a key role in influencing adoption decisions of merchants. In this context, the lack of transparency observed in the present study may reduce trust among merchants and discourage them from relying on fintech lending, leading them to prefer informal borrowing methods in certain situations.

11. Recommendations

Based on the findings of the study, the following recommendations are proposed to improve the effectiveness and adoption of fintech lending among micro-retailers:

1. Improving Financial Literacy among Micro-Retailers

The lack of financial and digital literacy among merchants is significant since they cannot comprehend concepts such as loan terms, interest rates, payment methods, and fees associated with the transactions. This could be addressed by initiating educational campaigns targeting the merchants and raising their awareness. In addition, the terms of the loan should be written in their local languages for easy comprehension.

2. Enhancing Awareness and Accessibility of Government Schemes

There is a need to improve awareness regarding government credit schemes such as PM Mudra Yojana, PM SVANidhi Scheme, and Stand-Up India Scheme. Outreach programs, simplified communication strategies, and local-level awareness campaigns should be strengthened. Additionally, simplifying application procedures can help small retailers access formal credit more easily.

3. Leveraging AI for Government Scheme Awareness

AI-enabled platforms can provide micro-retailers with information about relevant government schemes, including their eligibility criteria, benefits, and application procedures. Platforms such as Axis My India can help users identify suitable schemes and understand how to avail them. Such platforms should be promoted among small merchants to improve their awareness and access to available government support.

4. Improving Transparency in Digital Lending

It is important for fintech companies to clarify the terms of loans, such as what the differences are between approved funds and available funds, interest calculation, and fees. Transparency leads to clarity and trust between parties (Sulastri et al., 2024).

5. Enhancing Awareness of Government Credit Schemes

To improve awareness of government credit schemes, banks can set up a dedicated help desk or assign a staff member to guide merchants and other customers about available schemes. Since banks are commonly visited by all individuals, whether merchants or consumers, for financial purposes, this can serve as an effective way to increase awareness and help merchants to better understand and access such government schemes.

6. Building Trust in Fintech Platforms

Fintech platforms must foster trust among micro-retailers through clear communication, good customer service, and ethical business practices. These measures will decrease their reluctance and encourage micro-retailers to utilize fintech loan products.

12. Conclusion

This research evaluates QR-based fintech lending among micro-retailers in Delhi within the broader MSME sector. Despite many merchant respondents having received at least one loan offer via fintech, the majority did not accept or utilize these types of funding sources. The respondents identified various reasons for reluctance to accept or use these loans. Though fintech has enhanced accessibility to collateral-free lending, it has not yet achieved the same success in actual usage.

According to the research, there continues to exist a breach between the number of loans available within the loan providers and how many of these merchants apply for these loans.

This likely has a number of causes; for example, most merchants feel that interest rates charged through lending institutions are higher than what they would receive from an informal source (and that they have more faith in an informal lender), and many do not have sufficient information about the terms of loans available to them - nor are they very financially literate - making it difficult for merchants to decide whether or not to obtain a loan through either source. Additionally, while many merchants want to capitalize on the various programs offered by their respective governments, due to limited understanding and/or awareness, they are unable to take advantage of them.

During the field study, the field study indicated that fintech loans are mostly used for short-term financial needs instead of long-term business growth. Merchants generally prefer these loans for managing urgent or temporary business expenses. For larger or long-term financial requirements, they usually depend on other sources that they trust more. Fintech loans can help businesses grow only when the money is properly used for business purposes and not for personal expenses.

The disparity between the approved loan amount and the actual amount received by retailers was an additional point raised by participants in the research. A number of participants noted that they do not fully understand the reasons behind these deductions, and that the lenders failed to communicate effectively during the loan process. This contributes to uncertainty and reduces their trust regarding applying for loans and making financial choices.

Fintech lending provides a significant opportunity for improving the financial inclusion of small retailers that typically have difficulty obtaining formal credit through banks and other financial institutions. However, there are still practical issues and borrower behaviour that limit the effectiveness of fintech lending. Many merchants do not clearly understand the repayment terms, fees, and loans typically offered by fintech lenders, and feel uncomfortable using these services, thereby making it difficult for them to access credit. Additionally, issues related to a lack of transparency and financial literacy amplify merchants' uncertainty regarding using fintech lending products.

To enhance the utility and effectiveness of fintech lending, it will be necessary for fintech lenders to communicate effectively, educate potential borrowers about their products, and create collaborative, transparent, and trusting relationships with merchants. If these issues are resolved, fintech lending could extend beyond being a source of credit to serve as an effective infrastructure for stability and growth for micro-retailers within the MSME sector.

13. Appendices

Appendices: Questionnaire used for field survey

Title: Questionnaire on QR-Based Fintech Lending among Micro-Retailers in the MSME Sector

1. Duration of QR usage in business

□ Less than 6 months

□ 6 months – 1 year

□ 1–3 years

□ More than 3 years

2. Have you taken a QR-based loan from any fintech platform?

□ Currently using

□ Used earlier

□ Offered but not taken

□ Never offered

3. Do loan offers increase when your QR sales increase?

□ Very fast

□ Somewhat fast

□ Slow

□ No relation

4. Interest rate charged on QR-based loans is

□ Very low

□ Reasonable

□ High

□ Very high

5. Have QR-based loans helped in business growth?

□ Yes, significantly

□ Yes, slightly

□ No impact

□ Caused problems

6. Main use of QR-based loans

□ Inventory

□ Business expansion

□ Working capital

□ Personal use

7. Awareness about PM MUDRA Yojana

□ Fully aware

□ Somewhat aware

□ Heard only

□ Not aware

8. Awareness about Digital MSME Scheme (website / digital support subsidy)

□ Fully aware

□ Somewhat aware

□ Heard only

□ Not aware

9. Awareness of RBI rules for digital / fintech lending

□ Fully aware

□ Somewhat aware

□ Heard only

□ Not aware

10. Experience of fraud or hidden charges in QR-based loans

□ Faced personally

□ Heard from others

□ Suspected

□ Never

11. Understanding of loan terms (interest, EMI, penalties)

□ Fully confident

□ Somewhat confident

□ Low confidence

□ Not confident

12. Comparison of loan options before borrowing

□ Always

□ Sometimes

□ Rarely

□ Never

14. References

Acharya, M. (2025, September 2). *MSME Loan Scheme 2026 - Interest rate, how to apply, eligibility & types*. Cleartax. https://cleartax.in/s/msme-loan-scheme-2020

Ameriya, S., & Vyas, A. (2025). *Bharat PE and the Transformation of India's Digital payment ecosystem: a conceptual study*. https://www.researchgate.net/publication/393924362_Bharat_Pe_and_the_Transformation_of_India's_Digital_Payment_Ecosystem_A_Conceptual_Study

Anshika, Singla, A., & Mallik, G. (2021). Determinants of financial literacy: Empirical evidence from micro and small enterprises in India. *Asia Pacific Management Review, 26*(4), 248–255. https://doi.org/10.1016/j.apmrv.2021.03.001

Bakhshi, P., Agrawal, R. A., Mendon, S., Frank, D., Spulbar, C., Birau, R., & Filip, R. D. (2024). BARRIERS IN ADOPTION OF FINTECH BY STREET VENDORS AND HAWKERS IN INDIA USING INTERPRETIVE STRUCTURAL MODELING. *Verslas Teorija Ir Praktika*, *25*(1), 231–240. https://doi.org/10.3846/btp.2024.19208

Bhandari, N. (2026, January 9). *Understanding MSME Loan Interest Rates: Calculation Methods & Latest Rates (2026)*. FlexiLoans Finance, Business Loan Blogs, Tips & Guide. https://flexiloans.com/blog/msme-loan-interest-rates/

Cahyasari, D., Hamidah, H., & Widyastuti, U. (2025). Digitalization of Micro-Sized Enterprises: Determinants of Quick Response Code Indonesian Standard (QRIS), fintech usage behavior, and implications for business sustainability. *ABAC Journal*, *45*(3), 114–137. https://doi.org/10.59865/abacj.2025.18

Client challenge. (n.d.). Retrieved September 11, 2026, from https://www.scribd.com/document/725404465/Key-Facts-Statement-4666836?utm

Corporate profile. (2022). [Slide show]. Bharatpe. https://bharatpe.com/wp-content/uploads/2023/05/BharatPe-Corporate-Profile.pdf?utm

Dangkeng, A., & Munir, R. (2025). Fintech and Financial Inclusion: How digital payment systems Empower small entrepreneurs. *Journal of Indonesian Scholars for Social Research*, *5*(1), 46–55. https://doi.org/10.59065/jissr.v5i1.172

Fedbank Financial Services Limited. (n.d.). *Fedbank Financial Services Limited Annual report* (pp. 02–261). https://www.fedfina.com/site/assets/files/480398/annual_report_2023-24.pdf?utm_

Financial Literacy | RBL Bank. (n.d.). Retrieved September 11, 2026, from https://www.rbl.bank.in/financial-literacy

Ghosh, P., Vallee, B., & Zeng, Y. (2025). *FinTech lending and cashless payments*. https://doi.org/10.3386/w34148

Gugulothu, B., & Aleti, P. (2024). Contribution of MSMEs to entrepreneurial development

and National Growth in India: A Comprehensive analysis (2018–2023). *International Journal of Scientific Research and Modern Technology.*, 153. https://doi.org/10.38124/ijsrmt.v3i6.1210

Haris, C. (2024). Exploring factors and impacts of online lending on UMKM expansion in North Sulawesi. *JURNAL AL-QARDH*, *9*(1), 12–24. https://doi.org/10.23971/jaq.v9i1.8453

Kraiwanit, T., Pimsen, C., Limna, P., Jangjarat, K., & Moolngearn, P. (2025). Digital lending strategy in an emerging economy. *Corporate and Business Strategy Review, 7*(1), 8. https://doi.org/10.22495/cbsrv7i1art1

Ledi, K. K., Ameza-Xemalordzo, E., Amoako, G. K., & Asamoah, B. (2023). Effect of QR code and mobile money on performance of SMEs in developing countries. The role of dynamic capabilities. *Cogent Business & Management*, *10*(2). https://doi.org/10.1080/23311975.2023.2238977

Malik, S., Singh, K. N., & Ehigiamusoe, G. (2024). Fintech for MSME and the Role of Financial Education for MSMEs in Optimizing their Return as Investors. In *Apple Academic Press eBooks* (pp. 173–191). https://doi.org/10.1201/9781003535133-9

Metr, D. P. (2024). *Advancing MSMES through Global Integration and Cross-Border Connectivity: Opportunities for QRIS-Based Payment Code Development*. Universitas Indonesia, Depok, Indonesia. https://edunity.publikasikupublisher.com/index.php/Edunity/article/view/221/379

Mookerjee, J., Chattopadhyay, S., Ahmed, T., & Addepalli, L. (2022). Impact of QR-Codes as a disruptive technology during the COVId-19 contagion. *International Journal on Recent and Innovation Trends in Computing and Communication*, *10*(1s), 284–289. https://doi.org/10.17762/ijritcc.v10i1s.5850

My Mudra Fincorp Limited. (n.d.). *Unsecured business loan interest rate in India 2026*. My Mudra. Retrieved September 11, 2026, from https://www.mymudra.com/blog/unsecured-business-loan-interest-rate

Nambiar, J. (2025). A Study of Performance of PMMY among MSMEs in Promoting Financial Inclusion. *Zenodo (CERN European Organization for Nuclear Research)*. https://doi.org/10.5281/zenodo.14908884

Omowole, B. M., Urefe, O., Mokogwu, C., & Ewim, S. E. (2024). Integrating fintech and innovation in microfinance: Transforming credit accessibility for small businesses. *International Journal of Frontline Research and Reviews*, *3*(1), 090–100. https://doi.org/10.56355/ijfrr.2024.3.1.0032

Paytm Annual Report 2025. (2025). [Slide show]. Paytm. https://paytm.com/document/ir/agm/fy/-25/Paytm_Annual_Report_2025.pdf?utm

Ramesh, S., Haldar, A., & Kamat, M. (2025). Bridging the digital divide at the bottom of the pyramid: UTAUT insights into payment technology adoption by street vendors. *Qualitative Research in Financial Markets*, 1–22. https://doi.org/10.1108/qrfm-10-2024-0288

Sanjana, S. (2025). Empowering Kirana Stores in Mangalore: The impact of digital lending platforms on financial growth and sustainability. *INTERANTIONAL JOURNAL OF SCIENTIFIC RESEARCH IN ENGINEERING AND MANAGEMENT*, *09*(12), 1–9. https://doi.org/10.55041/ijsrem55118

Sekhar, K. R., & Saheb, S. S. (2026). User perceptions of RBI-approved P2P digital lending apps: an NLP, machine learning, and deep learning approach. *Frontiers in Artificial Intelligence*, *8*, 1708080. https://doi.org/10.3389/frai.2025.1708080

Sharma, S. K., Ilavarasan, P. V., & Karanasios, S. (2023). Small businesses and FinTech: a systematic review and future directions. *Electronic Commerce Research*, *24*(1), 535–575. https://doi.org/10.1007/s10660-023-09705-5

Sharma, S., R, L. P., Yadav, M., & Manoj, A. (2025). THE EFFECT OF FINTECH ON MICRO ENTERPRISES WITH SPECIAL REFERENCE TO KOCHI CITY. In SNEHA ABRAHAM, *ST. TERESA'S COLLEGE (AUTONOMOUS), ERNAKULAM*. http://117.239.78.102:8080/jspui/bitstream/123456789/6826/1/Simran%20Sharma_%20Project%20%281%29.pdf

Shelly, R., Sharma, T., & Bawa, S. S. (2020). ROLE OF MICRO, SMALL AND MEDIUM ENTERPRISES IN INDIAN ECONOMY. *International Journal of Economics and Financial Issues*, *10*(5), 84–91. https://doi.org/10.32479/ijefi.10459

Singh, K. N., & Sachdev, N. (2023). Fintech Literacy and Lending among Micro Steel Enterprises in Punjab, India. *International Journal of Revenue Management*, *13*(1). https://doi.org/10.1504/ijrm.2023.10058956

Sulastri, R., Janssen, M., Van De Poel, I., & Ding, A. (2024). Transforming towards inclusion-by-design: Information system design principles shaping data-driven financial inclusiveness. *Government Information Quarterly*, *41*(4), 101979. https://doi.org/10.1016/j.giq.2024.101979

Sushmita. (2025). Financial inclusion under the Pradhan Mantri Mudra Yojana. *European Economic Letters (EEL)*, 153–166. https://doi.org/10.52783/eel.v15i1s.3291

Tata Capital. (2025). *Know more about Business Loans* [Video]. Tata Capital. https://www.tatacapital.com/business-loan/rates-and-charges.html

Vasudevan, A., Rani, P. J., Raja, N., Nedumaran, G., Arokiasamy, A. R. A., & Qian, C. (2025). Fintech for sustainable agriculture: insights from Tamil Nadu, India. *Frontiers in Sustainable Food Systems*, *9*. https://doi.org/10.3389/fsufs.2025.1614553

Vision of a USD 5 trillion Indian economy. (2018). [Press release]. https://www.pib.gov.in/Pressreleaseshare.aspx?PRID=1549454&reg=3&lang=2